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ONCOLOGY

Central subfield thickness predicts visual acuity outcomes in plaque-irradiated eyes with choroidal melanoma

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Abstract

Objective To determine the association between pre-operative central subfield thickness (CST) and post-radiotherapy visual acuity (VA), cystoid macular edema (CME), and intravitreal anti-vascular endothelial growth factor (VEGF) requirement.

Design Single-center retrospective study.

Participants Patients with plaque-irradiated extramacular choroidal melanoma treated between 11/11/2011 and 4/30/2021. Pre-operative CST difference between the affected and unaffected eye was used. Kaplan-Meier analysis and hazard ratios were calculated.

Results Of 85 patients, pre-operative CST was greater in the melanoma-affected eye (vs. fellow eye) by mean of 20.4 μm (median 14.0, range − 60.0–182.0). Greater CST at presentation (vs. fellow eye) was associated with larger tumor diameter ($p = 0.02$), greater tumor thickness ($p < 0.001$), and more frequent tumor-related Bruch's membrane rupture ($p = 0.006$). On univariate analysis of outcome data, greater CST at presentation (vs. fellow eye) was associated with higher 5-year risk (1.09 [1.02–1.17], $p = 0.02$) of VA 20/200 or worse and increased (1.10 [1.01–1.20], $p = 0.03$) likelihood for anti-VEGF injections after plaque irradiation. There was no significant association with CME. The association between CST and VA outcome remained significant on multivariate analysis accounting for impact of tumor thickness and radiation dose to optic disc, while tumor distance to fovea was the only significant factor on multivariate analysis for anti-VEGF injections.

Conclusion Greater CST at presentation (vs. fellow eye) was associated with worse VA outcome following plaque radiotherapy for choroidal melanoma. Large-sized tumors may contribute to a higher intraocular VEGF burden, potentially leading to greater preoperative CST, which correlates with poor VA outcome post-plaque radiotherapy.

Keywords Tumor · Uvea · Choroid · Melanoma · Plaque radiotherapy · Cystoid macular edema

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Key messages

What is known

- Radiation maculopathy is a common sequela after plaque radiotherapy, and macular edema on spectral domain optical coherence tomography (SD-OCT) is often one of the earliest signs.

What is new

- In patients with plaque-irradiated extramacular choroidal melanoma, initial presentation with greater preoperative CST in the affected eye compared to the unaffected fellow eye was associated with increased 5-year risk for poor visual acuity of 20/200 or worse.
- Association between greater preoperative CST and poor visual acuity outcome after plaque irradiation remained significant after multivariate analysis accounting for tumor thickness and radiation dose to optic disc.

Introduction/background

Plaque radiotherapy has become one of the most common treatment modalities for management of uveal melanoma since the Collaborative Ocular Melanoma Study (COMS) showed 90% tumor control and 88% globe salvage rates at 5 years in patients with medium-sized tumors [1–3]. However, plaque radiotherapy exposes the surrounding unaffected ocular tissues to high-dose radiation, leading to side effects that can result in long-term vision loss [4–7].

Radiation retinopathy manifests with sequelae including retinal ischemia and non-perfusion that can lead to neovascularization, vitreous hemorrhage, tractional retinal detachment, and macular edema [8, 9]. Specifically, radiation maculopathy can affect up to 61% of patients by 24 months after plaque radiotherapy, with macular edema on spectral domain optical coherence tomography (SD-OCT) often being one of the earliest signs [6]. Factors such as younger age, diabetes, hypertension, radiation dose, and proximity of the tumor margin to the optic disc or fovea are all risk factors for developing radiation retinopathy [6, 10–13].

Given the devastating vision loss from radiation retinopathy, studies have targeted therapies such as intravitreal anti-vascular endothelial growth factor (VEGF) for delaying the onset or reducing the clinical impact of radiation-related side effects [14, 15]. Hence, it is important to identify patients who may be at higher risk of radiation retinopathy-related vision loss, as they may be more likely to benefit from earlier therapy. Mashayekhi et al. studied factors that contributed to the development of post-plaque CME [16]. However, to our knowledge, prior studies have not involved detailed preoperative macular OCT analysis to investigate associations with visual acuity loss and requirement for intravitreal anti-VEGF injections after plaque radiotherapy. Herein, we investigate pre-operative SD-OCT central subfield thickness (CST) in the affected eye (vs. fellow eye) to determine whether pre-operative CST is associated with post-plaque visual acuity, CME, and the need for intravitreal anti-VEGF injections in patients treated with plaque radiotherapy for extramacular choroidal melanoma.

Method

This study complied with the Health Insurance Portability and Accountability Act (HIPAA) and adhered to the tenets of the Declaration of Helsinki. The Mayo Clinic Institutional Review Board deemed this study exempt as a retrospective chart review, and informed consent was obtained from all patients for inclusion in the IRB-approved Prospective Ocular Tumors Study (POTS) database. Additional retrospective chart review was performed on a subset of patients enrolled in the POTS at Mayo Clinic, Rochester, MN, between July 15, 2019, and April 30, 2021. Patients were included in the study if they were diagnosed with a primary choroidal melanoma not involving the macula and treated with plaque radiotherapy between November 10, 2011, and April 30, 2021. Exclusion criteria were as follows: diagnosis of iris melanoma or a non-melanoma choroidal tumor, tumor location partially or completely within the macula, tumor location abutting or involving the optic disc, lack of pre-operative SD-OCT, treatment of choroidal melanoma with a non-plaque modality, prior treatment with intravitreal anti-VEGF or steroid injections, follow-up after plaque radiotherapy shorter than 1 year, presence of subfoveal subretinal fluid at presentation, or primary tumor treatment administered at an outside institution.

All study patients underwent a comprehensive eye examination by an ocular oncologist with color fundus drawing, fundus photography, autofluorescence, A and B scan ultrasonography, fluorescein (FA) and indocyanine green angiography (ICGA), and SD-OCT as indicated. Plaque radiotherapy was performed under anesthesia using Iodine-125 (I-125) Collaborative Ocular Melanoma Study (COMS) plaques with tumor localization using transillumination and/or indirect ophthalmoscopy. Intraoperative ultrasonography was used to confirm proper plaque placement. Planned treatment duration was 94 h for all patients.

Medical records were reviewed retrospectively to obtain the following data for each patient: demographic information (date of birth, race, sex); past medical history (skin malignancy, smoking history, diabetes, hypertension, hyperlipidemia, and history of *BAP1* mutation); family history of cutaneous or choroidal melanoma; personal ocular history (previous intravitreal anti-VEGF injections, previous intravitreal steroid injections, history of diabetic retinopathy, age-related macular degeneration, or retina vein occlusion); date of choroidal melanoma diagnosis; baseline visual acuity (VA); pre-operative CST in affected eye with choroidal melanoma and fellow unaffected eye; tumor parameters (distance to optic disc, distance to fovea, thickness, largest basal diameter, tumor thickness, Bruch's membrane rupture, ciliary body or iris involvement, extraocular extension, RPE trough, tumor location, anterior and posterior tumor margins, tumor shape, presence of subretinal fluid, color, orange pigment, and AJCC8 classification stage); treatment parameters (date and type of treatment, total treatment duration, radiation point doses to the tumor apex, tumor base, optic disc, fovea, lens, lacrimal gland, sclera, and opposite retina); prescription dose rate; and outcomes (post-operative follow-up dates, VA, presence of SD-OCT evident CME, CST, and requirement and date of first intravitreal anti-VEGF injection at defined time points of post-operative 2–4 months, 6–8 months, 12 months, 18 months, 24 months, 3 years, 4 years, 5 years, and final follow-up visit). Of note, the point doses refer to the total dose at a reference point in the structure of interest as defined by Rivard et al. [17] VA was assessed during ocular oncology clinic visits with the best available prescription, and pinhole acuity was consistently assessed if visual acuity was worse than 20/20. VA data were standardized in a logMAR format to allow for statistical analysis. Qualitative visual acuity values were assigned numerical values for logMAR. CF was assigned logMAR = 2, HM logMAR = 3, LP logMAR = 3, and NLP logMAR = 4. Some patients did not have data available for each follow-up study time point and were excluded from analysis when data were unavailable. Patients were divided into three categories for further analysis based on the presenting CST difference between the affected eye and fellow eye (presenting pre-operative CST of the eye with choroidal melanoma undergoing plaque radiotherapy minus presenting pre-operative CST of the fellow unaffected eye): lowest 25% CST difference (< 2 μm), middle 50% CST difference (2–28 μm) and highest 25% CST difference (> 28 μm). Of note, the presence of CME at defined post-operative follow-up timepoints was determined by SD-OCT image interpretation by two authors (S.M.S. and L.B.T.), with any uncertainties reviewed by the senior author (L.A.D.). Given the lack of data to guide normal and abnormal between-eye variation vs. disease-related asymmetry in CST, these study groups were determined to be fair for data analysis based on a mathematical approach guided by our statisticians.

EyeDose, an open-source Matlab program developed by Deufel et al., was used to calculate radiation dose to the tumor apex, tumor base, optic disc, fovea, lens, lacrimal gland, sclera, and opposite retina [18]. The program utilizes Monte Carlo methods to calculate three-dimensional heterogeneity corrected dose distributions for I-125 COMS plaques and has been validated against published Monte Carlo results for eight different tumor positions.

Statistical analysis was performed using SAS version 9.4 (Cary, NC). Demographics, clinical features, treatment features, and outcomes were compared between the lowest 25%, middle 50%, or highest 25% CST difference groups using chi-square tests for categorical variables and the Kruskal-Wallis test for continuous variables. Kaplan-Meier survival estimates were used to estimate the follow-up risk of development of visual acuity ≤ 20/200, CME, or need for injections as a function of time from plaque radiotherapy treatment. The effects of individual variables on the development of each outcome event were analyzed using a series of Cox proportional hazards regressions. Subsequent multivariate models included variables that were significant on a univariate level ($p < 0.05$).

Results

There were 85 patients with choroidal melanoma treated with plaque radiotherapy who met inclusion criteria. Patients were divided into three groups based on the difference in CST between the affected eye undergoing plaque radiotherapy and the unaffected fellow eye at the time of presentation prior to treatment. There were 19 patients in the lowest 25% CST difference cohort, 45 patients in the middle 50% CST difference cohort, and 21 patients in the highest 25% CST difference cohort. Table 1 describes patient demographics, including medical, family, and ocular history. There were 35 (41.2%) males and 50 (58.8%) females with mean age of 60.9 ± 13.8 years. All 85 (100%) patients were of White race. A comparison (lowest 25% vs. middle 25% vs. thickest 25% CST difference) revealed no significant differences in baseline demographics.

Table 1 Pre-operative predictability of post-radiation cystoid macular edema in patients with plaque-irradiated choroidal melanoma. Demographic features

Demographic features	Lowest 25% CST difference (< 2 µm between eyes) $n = 19$	Middle 50% CST difference (2–28 µm between eyes), $n = 45$	Highest 25% CST difference (> 28 µm between eyes), $n = 21$	*p* values	Combined, $n = 85$ (100%)
Age (years) Mean, median (range)	64.9, 65.0 (50.0–85.0)	57.6, 59.0 (21.0–87.0)	64.3, 62.0 (33.0–82.0)	0.06	60.9, 62.0 (21.0–87.0)
Involved eye					
Right	9 (47.4%)	23 (51.1%)	14 (66.7%)	0.41	46 (54.1%)
Left	10 (52.6%)	22 (48.9%)	7 (33.3%)		39 (45.9%)
Race					
White	19 (100.0%)	45 (100.0%)	21 (100.0%)	--	85 (100%)
Sex					
Male	6 (31.6%)	16 (35.6%)	13 (61.9%)	0.09	35 (41.2%)
Female	13 (68.4%)	29 (64.4%)	8 (38.1%)		50 (58.8%)
Medical history					
Skin melanoma	2 (10.5%)	2 (4.4%)	2 (9.5%)	0.62	6 (7.1%)
Previous smoker	1 (5.3%)	2 (4.4%)	2 (9.5%)	0.83	5 (5.9%)
Diabetes	3 (15.8%)	3 (6.7%)	2 (9.5%)	0.42	8 (9.4%)
Hypertension	7 (36.8%)	23 (51.1%)	13 (61.9%)	0.30	43 (50.6%)
Hyperlipidemia	7 (36.8%)	19 (42.2%)	14 (66.7%)	0.13	40 (47.1%)
Germline *BAP1* mutation	0 (0.0%)	0 (0.0%)	0 (0.0%)	--	0 (0.0%)
Family History					
Skin melanoma	4 (21.1%)	9 (20.0%)	2 (9.5%)	0.56	15 (17.6%)
Choroidal melanoma	1 (5.3%)	3 (6.7%)	0 (0.0%)	0.66	4 (4.7%)
Ocular history in affected eye					
Diabetic retinopathy	0 (0.0%)	1 (2.2%)	0 (0.0%)	1.00	1 (1.2%)
Macular degeneration (wet or dry)	1 (5.3%)	3 (6.7%)	2 (9.5%)	0.86	6 (7.1%)
Retinal vein occlusion	0 (0.0%)	0 (0.0%)	0 (0.0%)	--	0 (0.0%)

Abbreviations: *CST*, central subfield thickness

Bold *p* values are statistically significant

Clinical features at presentation with choroidal melanoma are detailed in Table 2. A comparison (lowest 25% vs. middle 25% vs. highest 25% CST difference) revealed that patients with greater difference in presenting CST between the affected and fellow eye had larger mean tumor basal diameter (10.1 mm vs. 11.9 mm vs. 13.0 mm, $p = 0.02$), greater tumor thickness (2.7 mm vs. 3.7 mm vs. 5.0 mm, $p < 0.001$), and more frequent tumor-related Bruch's membrane rupture (0.0% vs. 6.7% vs. 28.6%, $p = 0.006$). Pre-operative visual acuity and tumor location were similar between groups.

Treatment features are listed in Table 3. A comparison (lowest 25% vs. middle 50% vs. highest 25% CST difference) revealed that patients with the lowest CST difference between the affected and fellow eye had higher radiation dose to tumor apex (107.0 Gy vs. 98.0 Gy vs. 88.5 Gy, $p = 0.01$) and faster radiation dose rate (1.1 Gy/h vs. 1.0 Gy/h vs. 0.9 Gy/h, $p = 0.01$). Patients with greater CST difference had higher total radiation dose to lacrimal gland (4.1 Gy vs. 5.0 Gy vs. 8.1 Gy, $p = 0.003$), sclera (193.8 Gy vs. 209.7 Gy vs. 258.6 Gy, $p = 0.01$), and opposite retina (4.9 Gy vs. 5.5 Gy vs. 7.6 Gy, $p = 0.001$). There were no significant differences in radiation dose to fovea or optic disc.

Table 4 describes clinical outcomes at defined follow-up time points of 2–4 months, 6–8 months, 12 months, 18 months, 24 months, 3 years, 4 years, 5 years, and final follow-up visit. Overall, the frequency of outcomes of interest increased with greater time from radiation treatment. A comparison (lowest 25% vs. middle 50% vs. highest 25% CST difference) revealed that patients with higher CST difference between affected and fellow eyes had worse mean post-operative logMAR visual acuity at post-operative 2–4 months (0.2 vs. 0.4 vs. 0.5, $p = 0.02$), 24 months (0.4 vs. 0.3 vs. 0.9, $p = 0.004$), 36 months (0.5 vs. 0.5 vs. 1.0, $p = 0.008$), and 48 months (0.7 vs. 0.4

Table 2 Pre-operative predictability of post-radiation cystoid macular edema in patients with plaque-irradiated choroidal melanoma. Clinical features at presentation

Clinical features	Lowest 25% CST difference (< 2 µm between eyes) $n = 19$	Middle 50% CST difference (2–28 µm between eyes), $n = 45$	Highest 25% CST difference (> 28 µm between eyes), $n = 21$	p values	Combined, $n = 85$ (100%)
Pre-operative visual acuity affected eye (logMAR) Mean, median (range)	0.1, 0.1 (0.0–0.2)	0.1, 0.0 (0.0–0.2)	0.1, 0.0 (0.0–0.5)	0.16	0.1, 0.0 (0.0–0.5)
Pre-operative visual acuity fellow eye (logMAR) Mean, median (range)	1.3, 0.0 (0.0–4.0)	0.0, 0.0 (0.0–0.1)	0.0, 0.0 (0.0–0.9)	0.18	0.0, 0.0 (0.0–0.9)
Pre-operative CST affected eye (um) Mean, median (range)	277, 275 (239–324)	283, 281 (244–343)	346, 325 (273–480)	**< 0.001**	293, 293 (272–316)
Pre-operative CST fellow eye (um) Mean, median (range)	284, 277 (239–355)	271, 267 (219–323)	284, 284 (221–329)	0.06	277, 276 (219–355)
CST difference between affected and fellow eye (um) Mean, median (range)	− 7.6, − 3.0 (− 60.0–1.0)	12.7, 14.0 (2.0–28.0)	62.5, 41.0 (29.0–182.0)	**< 0.001**	20.4, 14.0 (− 60.0–182.0)
Cystoid macular edema (CME) on pre-operative OCT					
Absent	19 (100.0%)	45 (100.0%)	19 (95.0%)	0.20	83 (98.8%)
Present (non-fovea-involving)	0 (0.0%)	0 (0.0%)	1 (5.0%)		1 (1.2%)
Tumor features					
Distance to optic disc (mm) Mean, median (range)	7.4, 8.0 (0.0–14.0)	7.6, 7.0 (0.0–17.0)	5.5, 6.0 (0.0–14.0)	0.18	7.1, 7.0 (0.0–17.0)
Distance to fovea (mm) Mean, median (range)	7.5, 7.0 (3.0–13.0)	8.0, 8.0 (3.0–18.0)	6.0, 5.0 (2.0–13.0)	0.13	7.4, 7.0 (2.0–18.0)
Largest basal diameter (mm) Mean, median (range)	10.1, 10.0 (3.0–15.0)	11.9, 12.0 (5.0–22.0)	13.0, 13.0 (8.0–20.0)	**0.02**	11.8, 12.0 (3.0–22.0)
Tumor thickness (mm) Mean, median (range)	2.5, 2.3 (0.8–4.6)	3.7, 2.9 (0.9–10.3)	5.0, 5.2 (1.8–8.2)	**< 0.001**	3.7, 3.2 (0.8–10.3)
Bruch's membrane rupture	0 (0.0%)	3 (6.7%)	6 (28.6%)	**0.006**	9 (10.6%)
Ciliary body involvement	3 (15.8%)	4 (8.9%)	1 (4.8%)	0.48	8 (9.4%)
Iris involvement	0 (0.0%)	0 (0.0%)	0 (0.0%)	--	0 (0.0%)
Extraocular extension	0 (0.0%)	0 (0.0%)	0 (0.0%)	--	0 (0.0%)
RPE trough	1 (5.3%)	2 (4.4%)	0 (0.0%)	0.59	3 (3.5%)
Quadrantic location					
Macula	0 (0.0%)	0 (0.0%)	0 (0.0%)	0.69	0 (0.0%)
Inferior	4 (21.1%)	12 (26.7%)	5 (23.8%)		21 (24.7%)
Nasal	2 (10.5%)	11 (24.4%)	6 (28.6%)		19 (22.4%)
Superior	8 (42.1%)	10 (22.2%)	5 (23.8%)		23 (27.1%)
Temporal	5 (26.3%)	12 (26.7%)	5 (23.8%)		22 (25.9%)

vs. 1.4, $p = 0.004$). While not statistically significant, a similar trend was noted for post-operative 6–8 months, 12 months, 18 months, and 60 months. There was no significant difference in the proportion of patients with CME between the three groups at any of the follow-up time points.

Table 2 (continued)

Clinical features	Lowest 25% CST difference (< 2 µm between eyes) $n = 19$	Middle 50% CST difference (2–28 µm between eyes), $n = 45$	Highest 25% CST difference (> 28 µm between eyes), $n = 21$	*p* values	Combined, $n = 85$ (100%)
Anterior tumor margin					
Macula	0 (0.0%)	0 (0.0%)	0 (0.0%)	0.39	0 (0.0%)
Ciliary body	2 (10.5%)	5 (11.1%)	1 (4.8%)		8 (9.4%)
Equator to ora serrata	8 (42.1%)	22 (48.9%)	6 (28.6%)		36 (42.4%)
Macula to equator	9 (47.4%)	18 (40.0%)	14 (66.7%)		41 (48.2%)
Posterior tumor margin					
Macula	0 (0.0%)	0 (0.0%)	0 (0.0%)	0.31	0 (0.0%)
Equator to ora serrata	4 (21.1%)	6 (13.3%)	1 (4.8%)		11 (12.9%)
Macula to equator	15 (78.9%)	39 (86.7%)	20 (95.2%)		74 (87.1%)
Ciliary body	0 (0.0%)	0 (0.0%)	0 (0.0%)		0 (0.0%)
Tumor configuration by ultrasound					
Dome	18 (94.7%)	40 (88.9%)	12 (57.1%)	**0.003**	70 (82.4%)
Flat	1 (5.3%)	0 (0.0%)	0 (0.0%)		1 (1.2%)
Bilobed or multilobed	0 (0.0%)	3 (6.7%)	3 (14.3%)		6 (7.1%)
Mushroom	0 (0.0%)	2 (4.4%)	6 (28.6%)		8 (9.4%)
Quadrants of subretinal fluid					
None	5 (26.3%)	7 (15.6%)	3 (14.3%)	0.24	15 (17.6%)
Subretinal cap over tumor	11 (57.9%)	24 (53.3%)	8 (38.1%)		43 (50.6%)
1 quadrant	3 (15.8%)	10 (22.2%)	5 (23.8%)		18 (21.2%)
2 quadrants	0 (0.0%)	4 (8.9%)	5 (23.8%)		9 (10.6%)
Color					
Amelanotic	2 (10.5%)	5 (11.1%)	3 (14.3%)	0.56	10 (11.8%)
Partially pigmented	1 (5.3%)	10 (22.2%)	4 (19.0%)		15 (17.6%)
Pigmented	16 (84.2%)	30 (66.7%)	14 (66.7%)		60 (70.6%)
Autofluorescence					
Orange pigment	7 (36.8%)	20 (44.4%)	9 (42.9%)	0.85	36 (42.4%)
AJCC8 classification stage					
I	8 (42.1%)	20 (44.4%)	2 (9.5%)	0.06	30 (35.3%)
IIA	8 (42.1%)	16 (35.6%)	10 (47.6%)		34 (40.0%)
IIB	2 (10.5%)	4 (8.9%)	8 (38.1%)		14 (16.5%)
IIIA	1 (5.3%)	4 (8.9%)	1 (4.8%)		6 (7.1%)
IIIB	0 (0.0%)	1 (2.2%)	0 (0.0%)		1 (1.2%)

Abbreviations: *CST*, central subfield thickness

Bold *p* values are statistically significant

Table 5 describes cumulative proportions and hazard ratios for outcomes of post-radiation poor visual acuity ≤ 20/200, CME, and intravitreal anti-VEGF injections. A comparison (lowest 25% vs. middle 50% vs. highest 25% CST difference) demonstrated that greater difference in CST between affected and fellow eye was associated with more frequent poor visual acuity outcome ≤ 20/200 at 5 years (12% vs. 25% vs. 72%, $p = 0.004$, Fig. 1A). Patients with CST difference in the highest 25% were 5.14 times more likely (CI 1.44–18.35, $p = 0.01$) to have visual acuity ≤ 20/200 compared to patients with presenting CST in the lowest 25% difference group (Fig. 1A). The frequency of CME at 5 years was greatest in the highest 25% CST difference group, but this was not statistically significant (77% vs. 67% vs. 79%, $p = 0.84$, Fig. 1B). The proportion of patients requiring intravitreal anti-VEGF injections at 5 years was also similar between groups (74% vs. 57% vs. 72%, $p = 0.49$, Fig. 1C).

Univariate Cox regression analysis for factors associated with post-radiation visual acuity ≤ 20/200, CME, or requirement for anti-VEGF injections at 5 years is detailed in Table 6. Factors (hazard ratio [95% confidence interval], *p* value) associated with visual acuity ≤ 20/200 at 5 years included greater difference in CST between affected

Table 3 Pre-operative predictability of post-radiation cystoid macular edema in patients with plaque-irradiated choroidal melanoma. Treatment features

Plaque radiotherapy treatment features	Lowest 25% CST difference (< 2 µm between eyes), *n* = 19	Middle 50% CST difference (2–28 µm between eyes), *n* = 45	Highest 25% CST difference (> 28 µm between eyes), *n* = 21	*p* values	Combined, *n* = 85 (100%)
Total treatment duration (hours) Mean, median (range)	94, 94 (94–94)	94, 94 (94–94)	94, 94 (94–94)	--	94, 94 (94–94)
Total dose (Gy)					
Apex Mean, median (range)	107.0, 106.3 (81.9–145.4)	98.0, 95.8 (37.8–148.4)	88.5, 86.8 (63.8–114.0)	**0.01**	96, 92 (37.8–150.4)
Base Mean, median (range)	166.7, 157.2 (147.9–222.5)	177.3, 168.6 (119.3–354.1)	206.8, 186.3 (121.5–402.7)	0.07	182.3, 170.4 (119.3–402.7)
Disc Mean, median (range)	71.7, 25.1 (8.1–270.6)	43.8, 18.9 (8.2–210.4)	79.8, 34.9 (8.4–325.3)	0.07	59.3, 22.1 (8.1–325.3)
Fovea Mean, median (range)	53.8, 22.2 (9.7–238.3)	38.5, 20.3 (6.9–180.5)	67.4, 36.0 (12.2–283.5)	0.15	49.3, 22.8 (6.9–283.5)
Lens Mean, median (range)	11.7, 11.1 (7.4–26.0)	15.0, 13.0 (5.1–39.1)	21.8, 14.3 (7.1–91.3)	0.05	16.0, 13.0 (5.1–91.3)
Lacrimal gland Mean, median (range)	4.1, 4.7 (0.0–7.3)	5.0, 4.9 (0.0–12.1)	8.1, 7.7 (0.0–18.8)	**0.003**	5.6, 5.4 (0.0–18.8)
Sclera Mean, median (range)	193.8, 182.7 (164.–277.7)	209.7, 196.4 (134.0–430.1)	258.6, 232.1 (146.1–504.5)	**0.01**	218.4, 196.4 (134.0–504.5)
Opposite retina Mean, median (range)	4.9, 4.6 (4.0–6.5)	5.5, 5.0 (2.5–12.2)	7.6, 7.3 (3.7–14.3)	**0.001**	5.9, 5.0 (2.5–14.3)
Dose rate (Gy/h) Mean, median (range)	1.1, 1.1 (0.9–1.5)	1.0, 1.0 (0.4–1.6)	0.9, 0.9 (0.7–1.2)	**0.01**	1.0, 1.0 (0.4–1.6)

Abbreviations: *CST* central subfield thickness

Bold *p* values are statistically significant

Total dose represents the dose at a reference point in the structure of interest

and fellow eye (1.09 [CI 1.02–1.17], p = 0.02), greater pre-operative CST (1.08 [CI 1.02–1.15], p = 0.02), shorter tumor distance to fovea (1.14 [1.01–1.28], p = 0.04) and optic disc (1.10 [1.01–1.20], p = 0.03), greater tumor thickness (1.46 [CI 1.20–1.78], p < 0.001), and greater mean radiation dose to optic disc (1.01 [CI 1.00–1.02], p = 0.01). After adjustment for age and sex, tumor distance to fovea (1.14 [1.01–1.28], p = 0.04), tumor thickness (1.37 [CI 1.11–1.70], p = 0.004), and radiation dose to optic disc (1.01 [CI 1.00–1.02], p = 0.009) remained significant. No factors were significantly predictive of post-radiation CME risk in this series. Factors associated with need for anti-VEGF injections included greater difference in CST between affected and fellow eye (1.10 [CI 1.01–1.20], p = 0.03), greater pre-operative CST (1.07 [CI 1.00–1.14], p = 0.04), and shorter tumor distance from fovea (1.11 [CI 1.03–1.20], p = 0.01) and optic disc (1.08 [CI 1.01–1.15], p = 0.03). These factors remained significant after adjustment for age and sex.

Table 7 describes results of multivariate analysis using factors that were significant on univariate analysis. For risk of visual acuity ≤ 20/200, a model (HR [CI], p value) with pre-operative CST (1.09 [CI 1.02–1.02], p = 0.008), tumor thickness (1.29 [CI 1.05–1.59], p = 0.02), and radiation dose to optic disc (1.09 [CI 1.00–1.02], p = 0.04) showed all three variables to remain statistically significant. A second model with difference in CST between affected and fellow eye (1.11 [1.03–1.20], p = 0.005) and radiation dose to optic disc (1.01 [1.00–1.02], p = 0.002) also showed both variables to remain significant. For anti-VEGF injection requirement, a model with pre-operative CST (1.06 [0.99–1.13], p = 0.11) and tumor distance to fovea (0.90 [0.83–0.98], p = 0.02) showed that only tumor distance to fovea remained significant. A second model with difference in CST between affected and fellow eye (1.08 [0.99–1.18], p = 0.08) and tumor distance to fovea (0.91 [0.84–0.99], p = 0.02) also showed that only tumor distance to fovea remained significant.

Discussion/conclusion

This study investigated the association between greater presenting CST difference and outcomes of visual acuity, CME, and need to initiate anti-VEGF injections in patients undergoing plaque radiotherapy for extramacular choroidal melanoma. Patients with greater presenting CST in the affected

Table 4 Pre-operative predictability of post-radiation cystoid macular edema in patients with plaque-irradiated choroidal melanoma. Clinical outcomes at defined time points

Outcomes	Lowest 25% CST difference (< 2 μm between eyes), *n* = 19	Middle 50% CST difference (2–28 μm between eyes), *n* = 45	Highest 25% CST difference (> 28 μm between eyes), *n* = 21	*p* values	Combined, *n* = 85 (100%)
Post-operative 2–4 months					
Visual acuity affected eye (logMAR) *N* Mean, median (range)	17 0.2, 0.1 (0.0–1.7)	43 0.4, 0.1 (0.0–4.0)	19 0.5, 0.3 (0.0–3.0)	**0.02**	79 0.4, 0.2 (0.0–4.0)
Visual acuity unaffected eye (LogMAR) *N* Mean, median (range)	17 0.0, 0.0 (0.0–0.1)	43 0.1, 0.0 (0.0–6.0)	19 0.0, 0.0 (0.0–0.3)	0.17	79 0.0, 0.0 (0.0–0.6)
Cystoid macular edema (CME) on OCT	0/7 (0.0%)	2/11 (18.2%)	1/9 (11.1%)	0.76	3/27 (11.1%)
CST affected eye (um) *N* Mean, median (range)	7 281.6, 278.0 (264.0–315.0)	11 295.7, 299.0 (259.0–367.0)	8 352.9, 340.5 (262.0–483.0)	0.19	26 309.5, 286.0 (259.0–483.0)
CST unaffected eye (um) *N* Mean, median (range)	7 284.1, 285.0 (262.0–322.0)	13 287.9, 268.0 (263.0–288.0)	8 279.3, 285.0 (255.0–307.0)	0.07	28 256.5, 275.5 (255.0–322.0)
Post-operative 6–8 months					
Visual acuity affected eye (logMAR) *N* Mean, median (range)	15 0.2, 0.1 (0.0–1.4)	33 0.3, 0.1 (0.0–4.0)	13 0.3, 0.2 (0.0–2.0)	0.51	61 0.3, 0.1 (0.0–4.0)
Visual acuity unaffected eye (LogMAR) *N* Mean, median (range)	15, 0.0, 0.0 (0.0–0.1)	33, 0.0, 0.0 (0.0–0.4)	13, 0.0, 0.0 (0.0–0.1)	0.53	61, 0.0, 0.0 (0.0–0.4)
Cystoid macular edema (CME) on OCT	2/10 (20.0%)	3/23 (13.0%)	1/9 (11.1%)	0.85	6/42 (14.3%)
CST affected eye (um) *N* Mean, median (range)	9 314.1, 286.0 (258.0–537.0)	25 291.3, 283.0 (189.0–464.0)	9 335.1, 333.0 (275.0–393.0)	0.05	43 305.2, 287.0 (189.0–537.0)
CST unaffected eye (um) *N* Mean, median (range)	9 271.8, 268.0 (249.0–292.0)	25 274.8, 273.0 (236.0–317.0)	8 283.6, 285.0 (258.0–298.0)	0.24	43 275.9, 274.0 (236.0–317.0)
Post-operative 12 months					
Visual acuity affected eye (logMAR) *N* Mean, median (range)	15 0.3, 0.2 (0.0–1.5)	35 0.4, 0.1 (0.0–3.0)	17 0.5, 0.3 (0.0–3.0)	0.10	67 0.4, 0.2 (0.0–3.0)
Visual acuity unaffected eye (LogMAR) *N* Mean, median (range)	15 0.2, 0.0 (0.0–3.0)	35 0.2, 0.0 (0.0–3.0)	17 0.1, 0.0 (0.0–1.0)	0.41	67 0.2, 0.0 (0.0–3.0)
Cystoid macular edema (CME) on OCT	3/13 (23.1%)	9/27 (33.3%)	2/13 (15.4%)	0.52	14/53 (26.4%)
CST affected eye (um) *N* Mean, median (range)	14 331.9, 290.0 (250.0–785.0)	30 285.7, 279.5 (187.0–453.0)	14 330.4, 298.5 (226.0–533.0)	0.12	58 307.7, 286.0 (187.0–785.0)
CST unaffected eye (um) *N* Mean, median (range)	14 277.4, 278.0 (238.0–316.0)	28 266.7, 268.5 (233.0–312.0)	14 278.8, 287.0 (239.0–298.0)	**0.03**	56 272.4, 270.5 (233.0–316.0)

Table 4 (continued)

Outcomes	Lowest 25% CST difference (< 2 µm between eyes), *n* = 19	Middle 50% CST difference (2–28 µm between eyes), *n* = 45	Highest 25% CST difference (> 28 µm between eyes), *n* = 21	*p* values	Combined, *n* = 85 (100%)
Post-operative 18 months					
Visual acuity affected eye (logMAR) *N* Mean, median (range)	10 0.3, 0.2 (0.0–1.6)	30 0.7, 0.4 (0.0–4.0)	10 0.6, 0.4 (0.0–3.0)	0.40	50 0.6, 0.3 (0.0–4.0)
Visual acuity unaffected eye (LogMAR) *N* Mean, median (range)	10 0.0, 0.0 (0.0–0.0)	30 0.0, 0.0 (0.0–0.4)	10 0.1, 0.1 (0.0–0.3)	**0.02**	50 0.0, 0.0 (0.0–0.4)
Cystoid macular edema (CME) on OCT	5/9 (55.6%)	13/26 (50.0%)	3/7 (42.9%)	1.00	21/42 (50.0%)
CST affected eye (um) *N* Mean, median (range)	9 347.3, 309.0 (250.0–697.0)	26 10.2, 285.0 (198.0–576.0)	7 362.7, 336.0 (300–0–505.0)	0.05	42 326.9, 295.0 (198.0–697.0)
CSTs unaffected eye (um) *N* Mean, median (range)	9 281.8, 292.0 (244.0–305.0)	27 272.1, 271.0 (236.0–320.0)	7 285.3, 286.0 (268.0–301.0)	0.11	43 276.3, 278.0 (236.0–320.0)
Post-operative 24 months					
Visual acuity affected eye (logMAR) *N* Mean, median (range)	13 0.4, 0.2 (0.0–2.0)	33 0.3, 0.1 (0.0–3.0)	17 0.9, 0.5 (0.0–5.0)	**0.004**	63 0.5, 0.2 (0.0–5.0)
Visual acuity unaffected eye (LogMAR) *N* Mean, median (range)	13 0.0, 0.0 (0.0–0.1)	33 0.1, 0.0 (0.0–1.0)	17 0.1, 0.0 (0.0–0.3)	0.36	63 0.1, 0.0 (0.0–1.0)
Cystoid macular edema (CME) on OCT	6/12 (50.0%)	13/32 (40.6%)	9/14 (64.3%)	0.34	28/58 (48.3%)
CST affected eye (um) *N* Mean, median (range)	12 306.8, 291.0 (249.0–457.0)	32 301.0, 278.5 (208.0–552.0)	14 352.9, 302.0 (231.0–906.0)	0.21	58 314.7, 289.5 (208.0–906.0)
CST unaffected eye (um) *N* Mean, median (range)	12 272.1, 271.0 (239.0–296.0)	31 271.4, 268.0 (234.0–323.0)	15 283.2, 290.0 (243.0–300.0)	0.12	58 274.6, 272.5 (234.0–323.0)
Post-operative 36 months					
Visual acuity affected eye (logMAR) *N* Mean, median (range)	13 0.5, 0.3 (0.0–3.0)	33 0.5, 0.2 (0.0–3.0)	14 1.0, 0.8 (0.1–2.0)	**0.008**	60 0.6, 0.3 (0.0–3.0)
Visual acuity unaffected eye (LogMAR) *N* Mean, median (range)	13 0.0, 0.0 (0.0–0.2)	33 0.1, 0.0 (0.0–2.0)	15 0.0, 0.0 (0.0–0.2)	0.47	61 0.1, 0.0 (0.0–2.0)
Cystoid macular edema (CME) on OCT	6/11 (54.5%)	13/25 (46.4%)	11/13 (84.6%)	0.06	30/52 (57.7%)
CST affected eye (um) *N* Mean, median (range)	11 337.6, 291.0 (252.0–609.0)	28 310.8, 294.5 (175.0–827.0)	13 403.6, 302.0 (239.0–1161.0)	0.57	52 339.7, 294.5 (175.0–1161.0)
CST unaffected eye (um) *N* Mean, median (range)	11 268.1, 267.0 (236.0–302.0)	28 276.8, 275.5 (220.0–321.0)	13 272.7, 287.0 (143.0–299.0)	0.31	5 273.9, 275.0 (143.0–321.0)

Table 4 (continued)

Outcomes	Lowest 25% CST difference (< 2 µm between eyes), *n* = 19	Middle 50% CST difference (2–28 µm between eyes), *n* = 45	Highest 25% CST difference (> 28 µm between eyes), *n* = 21	*p* values	Combined, *n* = 85 (100%)
Post-operative 48 months					
Visual acuity affected eye (logMAR) *N* Mean, median (range)	11 0.7, 0.4 (0.0–3.0)	23 0.4, 0.2 (0.0–2.0)	13 1.4, 1.3 (0.0–3.0)	**0.004**	47 0.7, 0.4 (0.0–3.0)
Visual acuity unaffected eye (LogMAR) *N* Mean, median (range)	11 0.0, 0.0 (0.0–0.1)	23 0.0, 0.0 (0.0–0.3)	13 0.1, 0.0 (0.0–0.2)	0.22	47 0.0, 0.0 (0.0–0.3)
Cystoid macular edema (CME) on OCT	9/10 (90.0%)	12/22 (54.5%)	7/11 (63.6%)	0.16	28/43 (65.1%)
CST affected eye (um) *N* Mean, median (range)	10 371.8, 323.5 (246.0–609.0)	22 336.7, 301.5 (222.0–598.0)	10 364.0, 327.5 (240.0–619.0)	0.66	42 351.6, 315.0 (222.0–619.0)
CST unaffected eye (um) *N* Mean, median (range)	10 273.1, 267.0 (245.0–306.0)	21 271.0, 272.0 (222.0–323.0)	11 276.1, 279.0 (250.0–298.0)	0.89	42 272.9, 272.0 (222.0–323.0)
Post-operative 60 months					
Visual acuity affected eye (logMAR) *N* Mean, median (range)	9 0.7, 0.4 (0.0–3.0)	22 0.9, 0.4 (0.0–4.0)	12 1.0, 0.5 (0.1–1.8)	0.07	43 1.4, 0.6 (0.0–18.0)
Visual acuity unaffected eye (LogMAR) *N* Mean, median (range)	9 0.1, 0.0 (0.0–0.4)	22 0.0, 0.0 (0.0–0.2)	12 0.1, 0.0 (0.0–0.5)	0.09	43 0.1, 0.0 (0.0–0.5)
Cystoid macular edema (CME) on OCT	8/9 (88.9%)	13/22 (59.1%)	8/9 (88.9%)	0.18	29/40 (72.5%)
CST affected eye (um) *N* Mean, median (range)	9 459.1, 349.0 (255.0–952.0)	22 352.1, 319.5 (203.0–803.0)	9 364.0, 312.0 (215.0–931.0)	0.28	40 378.9, 326.0 (203.0–952.0)
CST unaffected eye (um) *N* Mean, median (range)	9 272.6, 267.0 (240.0–300.0)	22 275.9, 275.5 (221.0–321.0)	9 291.6, 286.0 (259.0–386.0)	0.64	40 278.7, 276.0 (221.0–386.0)
Final follow-up visit					
Follow-up time (years) Mean, median (range)	4.8, 5.1 (0.9–9.4)	5.4, 5.1 (1.0–12.5)	5.8, 5.8 (1.0–11.5)	0.62	5.4, 5.2 (0.9–12.5)
Enucleation	0 (0.0%)	0 (0.0%)	3 (14.3%)	**0.02**	3 (3.5%)
Visual acuity affected eye (LogMAR) Mean, median (range)	0.8, 0.3 (0.0–4.0)	0.7, 0.3 (0.0–4.0)	2.2, 1.3 (0.0–18.0)	**0.02**	1.1, 0.4 (0.0–18.0)
CST affected eye (um) Mean, median (range)	396.4, 326.0 (243.0–952.0)	331.8, 275.5 (175.0–1065.0)	364.1, 295.0 (166.0–837.0)	0.08	354.7, 289.0 (166.0–1065.0)
Time to enucleation (years) Mean, median (range)	NA	NA	3.1, 3.3 (2.1–3.8)	--	3.1, 3.3 (2.1–3.8)
Systemic outcomes					
Melanoma metastases					
Liver	0 (0.0%)	2 (%)	3 (%)	1.00	5 (%)
Lung	0 (0.0%)	0 (0.0%)	0 (0.0%)	--	0 (0.0%)
Bone	0 (0.0%)	1 (%)	0 (0.0%)	--	1 (%)

Abbreviations: *CST* central subfield thickness, *OCT* ocular coherence tomography

Bold *p* values are statistically significant

Table 5 Pre-operative predictability of post-radiation cystoid macular edema in patients with plaque-irradiated choroidal melanoma. Kaplan-Meier outcomes for visual acuity < 20/200, OCT-evident CME development, and requirement for intravitreal anti-VEGF injections

Outcomes	Lowest 25% CST difference (< 2 μm between eyes), *n* = 19	Middle 50% CST difference (2-–28 μm between eyes), *n* = 45	Highest 25% CST difference (> 28 μm between eyes), *n* = 21	*p* value at 60 months
	Cumulative proportion	Cumulative proportion	Cumulative proportion	
Visual acuity ≤ 20/200				
2–4 months	0%	0%	0%	**0.004**
6–8 months	5%	7%	10%	
12 months	5%	9%	10%	
24 months	12%	14%	10%	
36 months	12%	17%	32%	
48 months	12%	25%	44%	
60 months	12%	25%	72%	
Hazard ratio vs. lowest CST (95% CI) at 60 months		1.81 (0.51–6.35)		0.36
Hazard ratio vs. lowest CST (95% CI) at 60 months			5.14 (1.44–18.35)	**0.01**
OCT-evident cystoid macular edema				
2–4 months	0%	0%	0%	0.84
6–8 months	5%	7%	5%	
12 months	11%	20%	10%	
24 months	40%	39%	35%	
36 months	55%	55%	56%	
48 months	55%	63%	72%	
60 months	77%	67%	79%	
Hazard ratio vs. thinnest CST (95% CI) at 60 months		0.85 (0.42–1.72)		0.66
Hazard ratio vs. thinnest CST (95% CI) at 60 months			1.00 (0.46–2.16)	0.99
Requirement for intravitreal anti-VEGF injections				
2–4 months	0%	0%	0%	0.49
6–8 months	5%	0%	5%	
12 months	5%	11%	10%	
24 months	39%	33%	45%	
36 months	39%	42%	61%	
48 months	65%	48%	66%	
60 months	74%	57%	72%	
Hazard ratio vs. thinnest CST (95% CI) at 60 months		0.69 (0.35–1.40)		0.31
Hazard ratio vs. thinnest CST (95% CI) at 60 months			0.95 (0.44–2.03)	0.89

Abbreviations: *CST*, central subfield thickness; *CI*, confidence interval; *OCT*, optical coherence tomography; *VEGF*, vascular endothelial growth factor

Bold *p* values are statistically significant

eye compared to the fellow eye had worse post-operative visual acuity and more frequent requirement for intravitreal anti-VEGF injections. Greater pre-operative CST remained a significant factor for visual acuity outcome in multivariate modeling. Additionally, greater difference in presenting CST between the affected and fellow eye was associated with presenting features of larger tumor basal diameter, greater tumor thickness, and more frequent tumor-related Bruch's membrane rupture.

Previous studies have assessed factors influencing visual acuity outcome after plaque radiotherapy for uveal melanoma. Pagliara et al. studied the risk of vision loss after Ruthenium-106 plaque in 152 patients with choroidal melanoma and found the risk for vision loss increased with larger

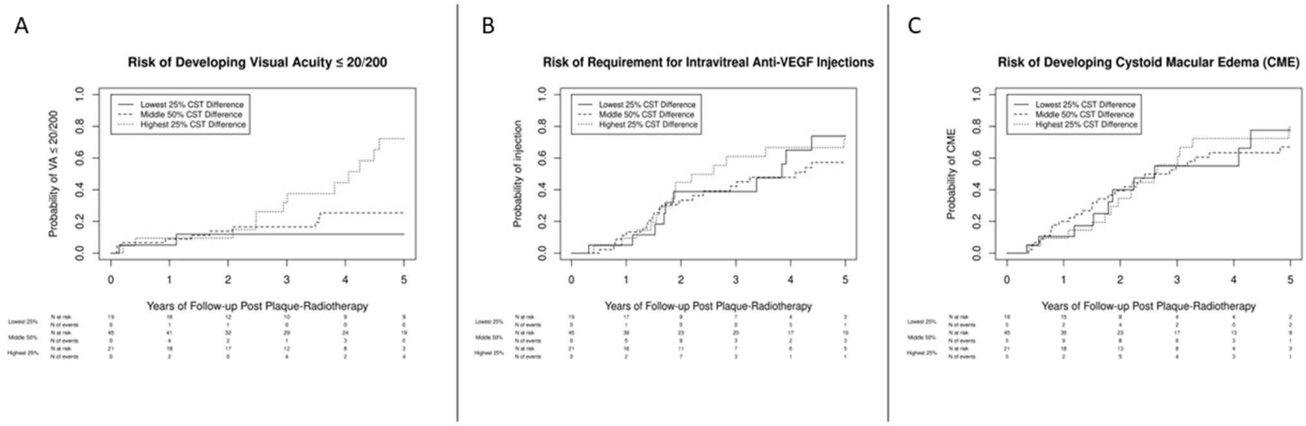

Fig. 1 Pre-operative central subfield thickness and post-plaque radiotherapy visual acuity, cystoid macular edema (CME), and anti-VEGF requirement for eyes with choroidal melanoma. Kaplan-Meier curves for 5-year outcomes after plaque radiotherapy in three groups defined by central subfield thickness (CST) difference between the affected and fellow eye in the lowest 25%, middle 50%, or highest 25%. **A** A comparison (lowest 25% vs. middle 25% vs. thickest 25% CST difference) revealed 12% vs. 25% vs. 72% (p = 0.004) probability of post-radiation visual acuity of 20/200 or worse at 5 years post-plaque radiotherapy. There was no difference between groups for 5-year risk of **B** CME (77% vs. 67% vs. 79%, p=0.84) or **C** treatment with intravitreal anti-vascular endothelial growth factor (VEGF) injections (74% vs. 57% vs. 72%, p = 0.49)

tumor basal diameter and posterior tumor margin closer to the foveola [19]. Damato et al. in a study of 458 patients treated with Ruthenium-106 brachytherapy reported vision loss worse than 20/200 was associated with reduced initial visual acuity and posterior tumor extension [20]. In a study of 227 patients treated with Ruthenium-106 plaque, Espensen et al. demonstrated increased risk of visual acuity loss with closer proximity between the tumor and optic disc and greater radiation dose to the optic disc [21]. In 2000, Shields et al. published an analysis of visual outcome in 1106 consecutive patients with uveal melanoma treated with I-125 plaque radiotherapy, and multivariable analysis revealed several factors associated with poor visual acuity (≤ 20/200), including poor initial visual acuity and greater tumor thickness [22]. Similarly, in the present study, greater tumor thickness, shorter tumor distance to fovea and optic disc, and greater radiation dose to optic disc were associated with increased risk of post-plaque visual acuity ≤ 20/200.

There are a limited number of studies that have assessed the relationship between CST and visual acuity specifically for uveal melanoma. A retrospective analysis of radiation maculopathy on OCT, OCT angiography, and fluorescein angiography in 93 patients treated with proton-beam irradiation for uveal melanoma by Matet et al. showed greater post-radiation central macular thickness (CMT) and absolute difference in post-radiation CMT of treated and fellow control eyes were associated with worse visual acuity outcomes (VA < 20/200) [23]. Horgan et al. studied risk factors for the development of macular edema following I-125 plaque radiotherapy and adjunctive transpupillary thermotherapy (TTT) laser in 135 patients with uveal melanoma and found a significant correlation between visual acuity and CST at onset of macular edema and maximal macular edema (rho = 0.516, p < 0.0001) [6]. These prior studies did not specifically assess the predictability of pre-operative CST on post-plaque visual acuity but seem to be in line with our finding that greater presenting CST in the affected eye confers a higher risk of poor post-plaque visual acuity outcome.

Early immunohistochemical studies by Stitt et al., Boyd et al., and Missotten et al. and a meta-analysis by Yang et al. have shown that the aqueous and vitreous of eyes with uveal melanoma express higher levels of VEGF compared to eyes without the tumor [24–26]. In Yang et al.'s meta-analysis, larger sized tumors showed a significantly greater level of VEGF expression [24]. This was also reported in a study by Parrozzani et al. [27]. A study by Chen et al. showed a positive correlation between tumor thickness and VEGF-A expression [28], and Missotten et al. showed a positive correlation between VEGF-A levels and larger tumor basal tumor and height [25]. Thus, tumors of larger size produce more VEGF, which could be responsible for the higher pre-operative CST in eyes with larger choroidal melanoma as seen in our study. In line with this, we found greater presenting tumor basal diameter and thickness were associated with greater pre-operative CST (vs. fellow eye). Perhaps tumor-associated VEGF production underlies the association between greater presenting CST and outcomes of worse visual acuity and anti-VEGF requirement post-plaque radiotherapy.

Cystoid macular edema is a well-recognized sequela of I-125 plaque radiotherapy for choroidal melanoma that has

Table 6 Pre-operative predictability of post-radiation cystoid macular edema in patients with plaque-irradiated choroidal melanoma. Univariate Cox-proportional hazard ratio model for post-radiation visual acuity ≤ 20/200, CME development, and need for anti-VEGF injections

	Hazard ratio (HR)	95% CI	*p* value	HR age/sex adjusted	95% CI	*p* value
Variables associated with post-radiation visual acuity ≤ 20/200						
Difference in CST (affected–fellow eye) (10-µm scale)	1.09	1.02–1.17	**0.02**	1.06	0.99–1.15	0.11
Pre-operative CST (10-µm scale)	1.08	1.02–1.15	**0.006**	1.06	1.00–1.13	0.052
Tumor distance to fovea* (mm)	1.14	1.01–1.28	**0.04**	1.14	1.01–1.28	**0.04**
Tumor distance to optic nerve* (mm)	1.10	1.01–1.20	**0.03**	1.08	0.98–1.18	0.11
Tumor thickness (mm)	1.46	1.20–1.78	**< 0.001**	1.37	1.11–1.70	**0.004**
Tumor basal diameter (mm)	1.06	0.94–1.19	0.37	1.00	0.89–1.13	1.00
Total dose to fovea	1.01	0.99–1.01	0.11	1.01	0.99–1.01	0.09
Total dose to disc	1.01	1.00–1.02	**0.006**	1.01	1.00–1.02	**0.009**
Variables associated with post-radiation CME						
Difference in CST (affected–fellow eye) (10-µm scale)	1.03	0.95–1.12	0.43	1.04	0.96–1.12	0.32
Pre-operative CST (10-µm scale)	1.01	0.95–1.08	0.68	1.03	0.97–1.09	0.35
Tumor distance to fovea* (mm)	0.93	0.86–1.01	0.07	0.94	0.86–1.00	0.05
Tumor distance to optic nerve* (mm)	0.96	0.91–1.02	0.16	0.95	0.89–1.01	0.09
Tumor thickness (mm)	1.01	0.88–1.16	0.90	1.01	0.87–1.17	0.88
Tumor basal diameter (mm)	0.95	0.87–1.04	0.29	0.95	0.86–1.05	0.29
Total dose to fovea	1.00	0.99–1.01	0.29	1.00	0.99–1.01	0.45
Total dose to disc	1.00	0.99–1.01	0.20	1.00	0.99–1.01	0.29
Variables associated with requirement for anti-VEGF injections						
Difference in CST (affected–fellow eye) (10-µm scale)	1.10	1.01–1.20	**0.03**	1.13	1.04–1.24	**0.006**
Pre-operative CST (10=µm scale)	1.07	1.00–1.14	**0.04**	1.10	1.03–1.18	**0.006**
Tumor distance to fovea* (mm)	1.11	1.03–1.20	**0.008**	1.12	1.03–1.22	**0.005**
Tumor distance to optic nerve* (mm)	1.08	1.01–1.15	**0.03**	1.09	1.02–1.16	**0.02**
Tumor thickness (mm)	0.93	0.80–1.08	0.34	0.89	0.75–1.06	0.18
Tumor basal diameter (mm)	0.95	0.86–1.04	0.28	0.94	0.85–1.04	0.21
Total dose to fovea	1.00	0.99–1.01	0.97	1.00	0.99–1.01	0.80
Total dose to disc	1.00	0.99–1.01	0.83	1.00	0.99–1.01	0.97

Abbreviations: *CST*, central subfield thickness; *CME*, cystoid macular edema

Total dose represents the dose at a reference point in the structure of interest

*The hazard ratios for the tumor distance to fovea and optic nerve are presented for a decrease in the variable, so that the possible increased risk with association of shorter distances between tumor and fovea/optic nerve can be assessed

a mean onset of 12 months after radiation but can occur as early as 4 months in some patients.[6] A cumulative incidence of CME as high as 40 to 70% has been reported in patients receiving plaque radiotherapy [6, 29–31]. In a study of 306 patients with uveal melanoma, Mashayekhi et al. noted a correlation between the presence of pre-plaque subclinical macular edema (defined as increased CST of > 10 µm compared with opposite eye) and greater presenting tumor thickness, distance to fovea, and risk of future CME after plaque radiotherapy [16]. Our study did not show an association between pre-operative CST, the difference in CST between the affected and fellow eye, or factors such as tumor thickness with post-operative CME development. However, greater CST difference at presentation and shorter distance to fovea were associated with a greater need for anti-VEGF injections. The lack of association between pre-plaque CST and post-plaque CME in our study could be due to the small sample size and frequent patient follow-up for radiation retinopathy outside of our institution, where CME could have been initially detected. The need for anti-VEGF could serve as a surrogate marker of CME, and the factors associated with need for anti-VEGF injections were similar in our study and the work by Mashayekhi et al. Still, CST did not remain significant on multivariate analysis for anti-VEGF requirement, indicating that other factors could be responsible for the association between presenting CST and VA outcome. We further acknowledge the complex relationship between VA, CME, and anti-VEGF treatment and note that 6-18 month and >48-month time points did not show differences in VA between CST groups. This could potentially be explained by a period of CME controlled by anti-VEGF

Table 7 Pre-operative predictability of post-radiation cystoid macular edema in patients with plaque-irradiated choroidal melanoma. Multivariate Cox-proportional hazard ratio model for post-radiation visual acuity ≤ 20/200 and need for anti-VEGF injections

Multivariate Cox-regression analysis			
Variables associated with post-radiation visual acuity ≤ 20/200	Hazard ratio	95% CI	*p* value
Model 1			
Pre-operative CST (10 um scale)	1.09	1.02–1.02	**0.008**
Tumor thickness	1.29	1.05–1.59	**0.02**
Total dose to disc	1.01	1.00–1.02	**0.04**
Model 2			
Difference in CST (affected–fellow eye) (10-µm scale)	1.11	1.03–1.20	**0.005**
Total dose to disc	1.01	1.00–1.02	**0.002**
Variables associated with need for anti-VEGF injections			
Model 1			
Tumor distance to fovea	0.90	0.83–0.98	**0.02**
Pre-operative CST (10-µm scale)	1.06	0.99–1.13	0.11
Model 2			
Tumor distance to fovea	0.91	0.84–0.99	**0.03**
Difference in CST (affected–fellow eye) (10-µm scale)	1.08	0.99–1.18	0.08

HR, hazard ratio; *CST*, central subfield thickness

Bold *p* values are statistically significant

Total dose represents the dose at a reference point in the structure of interest

treatment in a subset of patients with moderate intraocular VEGF burden, while patients with greater presenting CST (and potentially higher intraocular VEGF burden) could not be controlled despite injections. We may subsequently see a failure of anti-VEGF to treat other radiation side effects such as macular ischemia or optic atrophy after longer term follow-up, and such untreated side effects become equalizers in VA between CST groups by 5 years post-treatment.

To ensure that differences in radiation dose based on inherent differences in tumor size were not impacting our study findings, we performed multivariate analysis for visual acuity outcome along with CST in Model 1 (Table 7) utilizing tumor thickness as a surrogate variable for radiation dose. Tumor thickness and radiation dose to optic disc continued to remain significant on multivariate analysis for visual acuity outcome along with CST. CST difference remained significant for visual acuity outcome even with step-wise multivariate analysis inputting radiation parameters first, indicating an impact of CST beyond radiation dose effect. When considering anti-VEGF burden, distance to fovea remained significant on multivariate modeling, while CST did not, suggesting that factors other than CST, such as radiation dose, might be more important for prediction of anti-VEGF burden. We also acknowledge that our study showed a significant association between CST difference in affected vs. fellow eyes and irradiation dose to the lacrimal gland and opposite retina but did not show a correlation between presenting CST and radiation dose to fovea and optic disc, which may depend more on tumor location than on tumor volume alone.

Study limitations include small sample size and a retrospective study design, which limited our ability to obtain an updated refraction and best-corrected visual acuity at every visit. Additionally, the presence of CME at defined post-operative follow-up timepoints was determined by SD-OCT image interpretation by two authors (S.M.S. and L.B.T.) which could contribute to subjectivity. While we studied the presence or absence of CME, we did not report the time of first CME as an outcome, as sometimes this may have been captured outside of our facility. Non-uniform management of radiation maculopathy and retinopathy due to frequent shared care with local retina providers limited availability of detailed information such as type and frequency of anti-VEGF injections. Hence, the date of first intravitreal anti-VEGF was obtained as a more reliable datapoint. While it is likely that most intravitreal anti-VEGF injections were administered for treatment of post-plaque CME, the need for anti-VEGF injections was deemed by the patient's retina provider since this was a retrospective study. Also, this study was not designed to assess whether these injections were beneficial for visual acuity preservation. Finally, we understand that patients can have variation in baseline CST between eyes, with the Beaver Dam Eye Study showing a 0.89 inter-eye correlation in CST [32]. Given that there are no established cutoffs in the literature for normal variation in CST between eyes, we deemed the cutoffs in our study based on what seemed appropriate in statistical calculations and would allow detection of significant differences.

In summary, for plaque-irradiated extramacular choroidal melanoma, initial presentation with greater preoperative CST in the affected eye compared to the unaffected fellow eye was associated with increased 5-year risk for poor visual acuity of 20/200 or worse. Preoperative CST did not predict 5-year risk of post-radiation maculopathy as measured by presence of CME, but analysis was limited due to the non-uniform treatment with intravitreal anti-VEGF agents. Larger studies are needed to validate the predictive value of presenting CST for post-plaque visual acuity outcomes, and future studies should investigate whether patients with greater preoperative CST are more likely to benefit from earlier, more frequent, and/or prophylactic anti-VEGF injections.

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Declarations

Ethics approval The Mayo Clinic Institutional Review Board deemed this study exempt as a retrospective chart review, and informed consent was obtained from all patients for inclusion in the IRB-approved Prospective Ocular Tumors Study (POTS) database.

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